

Addendum for DPP 2016 February West Area Group Study

February 1, 2019

This addendum contains updated study results for the Report entitled *MISO DPP 2016 February West Area Phase 3 Study* and dated June 26, 2018. Results have been updated to reflect the revised PJM affected system study report and the Lathrop facilities study report.

The revised PJM affected system study report has two major changes:

- 1) Projects responsible for mitigating the Twin Branch–Argenta 345 kV line overload are J498, J499, J526, J530, instead of J498, J499, J526, J528 in the previous PJM affected system study report.
- 2) The Benton–T-094 345 kV is no longer a constraint. Therefore, projects in DPP 2016 February West are not responsible for mitigation.

The Lathrop facilities study report determined that the DPP 2016 February West group was not responsible for the upgrade of 25 MVAR switched cap bank at Lathrop 161 kV (\$2,000,000).

The revised per project cost summary of generation projects is listed below.

1.1.1 J432 Summary

Network Upgrade	Cost	J432	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$4,129	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$29,917	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$57,662	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$163,941	Reactive Power NU
Total Cost Per Project for ERIS for each Project		\$255,650	

1.1.2 J460 Summary

Network Upgrade	Cost	J460	NUs Type
Hazel Creek 345-230-13.8 kV xfmr #6	\$4,900,000	\$2,844,327	SH MISO
Hankinson-Wahpeton 230 kV	\$1,050,000	\$99,674	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$8,049	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$56,326	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$112,589	Reactive Power NU

Network Upgrade	Cost	J460	NUs Type
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$337,169	Reactive Power NU
Total Cost Per Project for ERIS for each Project		\$3,458,135	

1.1.3 J475 Summary

Network Upgrade	Cost	J475	NUs Type
J530 POI-Hills 345 kV	\$300,000	\$115,118	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$22,955	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$612,929	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$10,811	Reactive Power NU
Irvine-Powershiek 161 kV	\$11,000,000	\$11,000,000	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$11,761,813	

1.1.4 J485 Summary

Network Upgrade	Cost	J485	NUs Type
Not Required	\$0	\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.1.5 J488 Summary

Network Upgrade	Cost	J488	NUs Type
Big Stone 115-230-13.8 kV xfmr	\$3,250,000	\$785,521	SH MISO
Big Stone-Browns Valley 230 kV	\$350,000	\$78,318	SH MISO
Big Stone-Blair 230 kV	\$200,000	\$48,343	SH MISO
Hankinson-Wahpeton 230 kV	\$1,050,000	\$248,664	SH MISO
Hankinson-Forman 230 kV	\$400,000	\$400,000	SH MISO
Oakes-Forman 230 kV	\$950,000	\$950,000	SH MISO
Oakes-Ellendale 230 kV	\$1,650,000	\$1,650,000	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$6,072	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$42,960	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$84,860	Reactive Power NU

Network Upgrade	Cost	J488	NUs Type
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$245,043	Reactive Power NU
Total Cost Per Project for ERIS for each Project		\$4,539,782	

1.1.6 J493 Summary

Network Upgrade	Cost	J493	NUs Type
Johnson Jct-Ortonville 115 kV	\$16,850,000	\$5,603,545	SH MISO
Johnson Jct-Morris 115 kV	\$8,725,000	\$2,901,539	SH MISO
Big Stone 115-230-13.8 kV xfmr	\$3,250,000	\$821,453	SH MISO
Big Stone-Browns Valley 230 kV	\$350,000	\$90,564	SH MISO
Big Stone-Blair 230 kV	\$200,000	\$50,552	SH MISO
Hankinson-Wahpeton 230 kV	\$1,050,000	\$233,879	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$6,243	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$44,968	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$87,186	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$248,985	Reactive Power NU
Total Cost Per Project for ERIS for each Project		\$10,088,914	

1.1.7 J495 Summary

Network Upgrade	Cost	J495	NUs Type
Hazleton-Washburn 161 kV	\$16,700,000	\$16,700,000	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$7,422	SH MISO
Lundquist-Deere Northeast 161 kV	\$100,000	\$100,000	SH MISO
Deere Northeast-Washburn 161 kV	\$100,000	\$100,000	SH MISO
Liberty-Dundee 161 kV	\$5,000,000	\$3,538,324	SH CIPCO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$38,168	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$106,516	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$945,053	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$21,535,483	

1.1.8 J498 Summary

Network Upgrade	Cost	J498	NUs Type
Boone Jct-Sub T Fort Dodge 161 kV	\$7,000,000	\$7,000,000	SH MISO
Poweshiek-Reasnor 161 kV	\$290,000	\$76,715	SH MISO
Grimes-Sycamore 345 kV #2	\$1,200,000	\$519,976	SH MISO
Grimes-Beaver Creek 345 kV	\$600,000	\$318,003	SH MISO
Granger Tap-108th & 54th 161 kV	\$2,000,000	\$889,769	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$34,333	SH MISO
Franklin-Wall Lake 161 kV	\$200,000	\$135,553	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$56,844	PJM AFS
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$290,525	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$470,728	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$499,180	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$10,291,626	

1.1.9 J499 Summary

Network Upgrade	Cost	J499	NUs Type
Poweshiek-Reasnor 161 kV	\$290,000	\$88,935	SH MISO
Grimes-Sycamore 345 kV #2	\$1,200,000	\$680,024	SH MISO
Granger Tap-108th & 54th 161 kV	\$2,000,000	\$1,110,231	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$43,684	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$56,062	PJM AFS
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$372,767	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$578,935	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$234,601	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,165,239	

1.1.10 J500 Summary

Network Upgrade	Cost	J500	NUs Type
Poweshiek-Reasnor 161 kV	\$290,000	\$124,351	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$60,570	SH MISO

Network Upgrade	Cost	J500	NUs Type
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$533,626	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$802,791	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$311,486	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,832,823	

1.1.11 J504 Summary

Network Upgrade	Cost	J504	NUs Type
Mqoketa-Wyoming 161 kV	\$200,000	\$26,012	SH CIPCO
Liberty-Dundee 161 kV	\$5,000,000	\$797,016	SH CIPCO
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$8,546	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$5,646	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$837,220	

1.1.12 J506 Summary

Network Upgrade	Cost	J506	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$12,066	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$100,935	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$167,016	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$364,509	Reactive Power NU
Total Cost Per Project for ERIS for each Project		\$644,526	

1.1.13 J510 Summary

Network Upgrade	Cost	J510	NUs Type
Not Required		\$0	
Total Cost Per Project for ERIS for each Project		\$0	

1.1.14 J514 Summary

Network Upgrade	Cost	J514	NUs Type
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Network Upgrade	Cost	J514	NUs Type
Marshalltown-Blairstown 115 kV	\$600,000	\$600,000	SH MISO
Blairstown-Prairie Crk 115 kV	\$1,700,000	\$1,700,000	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$1,199	SH MISO
Mqoketa-Wyoming 161 kV	\$200,000	\$18,725	SH CIPCO
Liberty-Dundee 161 kV	\$5,000,000	\$664,660	SH CIPCO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$8,084	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$16,929	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$47,685	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,057,281	

1.1.15 J523 Summary

Network Upgrade	Cost	J523	NUs Type
Adams 345-161-13.8 kV xfmr	\$100,000	\$100,000	SH MISO
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$7,134	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$56,507	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$163,641	

1.1.16 J524 Summary

Network Upgrade	Cost	J524	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$21,174	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$3,437	SH MISO
Webster-Sub T Ft Dodge 161 kV	\$6,250,000	\$6,250,000	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$28,506	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$47,635	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$113,639	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$6,464,390	

1.1.17 J526 Summary

Network Upgrade	Cost	J526	NUs Type
Hazel Creek 345-230-13.8 kV xfmr #6	\$4,900,000	\$2,055,673	SH MISO
Panther-McLeod 230 kV	\$1,000,000	\$1,000,000	SH MISO
Johnson Jct-Ortonville 115 kV	\$16,850,000	\$11,246,455	SH MISO
Johnson Jct-Morris 115 kV	\$8,725,000	\$5,823,461	SH MISO
Big Stone 115-230-13.8 kV xfmr	\$3,250,000	\$1,643,026	SH MISO
Big Stone-Browns Valley 230 kV	\$350,000	\$181,117	SH MISO
Big Stone-Blair 230 kV	\$200,000	\$101,106	SH MISO
Hankinson-Wahpeton 230 kV	\$1,050,000	\$467,782	SH MISO
Wahpeton-Fergus Falls 230 kV	\$100,000	\$100,000	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$12,428	SH MISO
Twin Branch—Argenta 345 kV	\$208,000	\$51,564	PJM AFS
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$89,320	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$173,588	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$496,528	Reactive Power NU
Total Cost Per Project for ERIS for each Project		\$23,442,049	

1.1.18 J527 Summary

Network Upgrade	Cost	J527	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$30,834	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$270,924	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$408,583	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$155,623	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$865,964	

1.1.19 J528 Summary

Network Upgrade	Cost	J528	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$24,689	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$216,876	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$327,174	Reactive Power NU

Network Upgrade	Cost	J528	NUs Type
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$123,681	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$692,421	

1.1.20 J529 Summary

Network Upgrade	Cost	J529	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$13,533	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$100,644	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$188,821	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$640,330	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$943,327	

1.1.21 J530 Summary

Network Upgrade	Cost	J530	NUs Type
J530 POI-Hills 345 kV	\$300,000	\$184,882	SH MISO
Parnell-J438 POI 161 kV	\$100,000	\$100,000	SH MISO
Mqoketa-Wyoming 161 kV	\$200,000	\$155,263	SH CIPCO
Twin Branch—Argenta 345 kV	\$208,000	\$43,530	PJM AFS
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$307,033	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$790,708	

1.1.22 J534 Summary

Network Upgrade	Cost	J534	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$101,961	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$14,780	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$111,693	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$205,752	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$649,340	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,083,526	

1.1.23 J535 Summary

Network Upgrade	Cost	J535	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$158,862	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$16,532	SH MISO
Franklin-Wall Lake 161 kV	\$200,000	\$64,447	SH MISO
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$140,805	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$227,591	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$350,243	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$958,481	